Power

File: power.kicad_sch

Hub

File: hub.kicad_sch

FT1

File: ft1.kicad_sch

FT2

File: ft2.kicad_sch

FT3

File: ft3.kicad_sch

FT4

File: ft4.kicad_sch

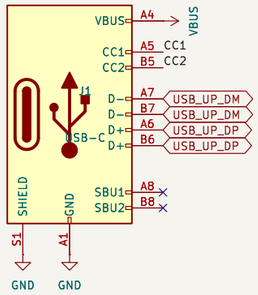
SBC farm

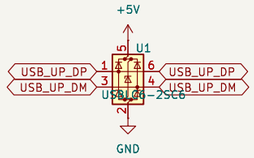
Sheet: /
File: sbc-uart-hub.kicad_sch

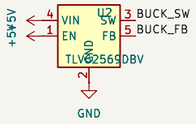
Title: SBC farm 16-ch UART hub

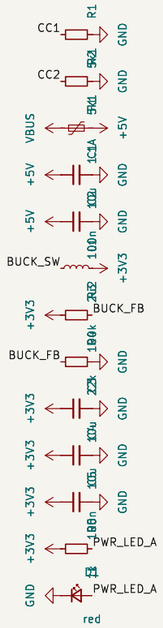
Size: A4	Date:	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 1/7

USB-C (USB 2.0 only). Bus powered, budget ~0.45 A @5 V via buck.













SBC farm

Sheet: /Power/
File: power.kicad_sch

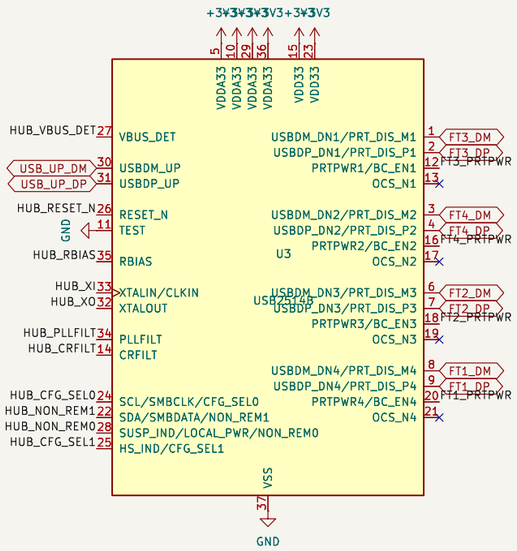
Title: USB-C input, 3.3 V buck

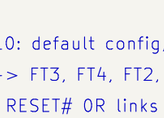
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 2/7

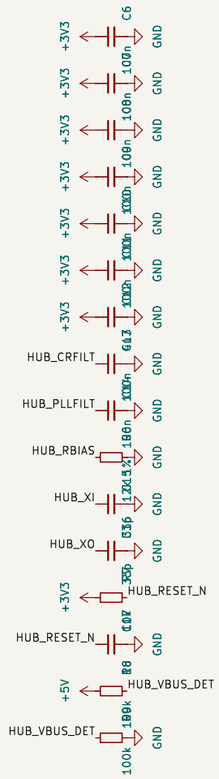
CFG_SEL[1:0]=10: default config, bus powered. NON_REM=00. BC_EN=0 (PRTPWR not pulled up).

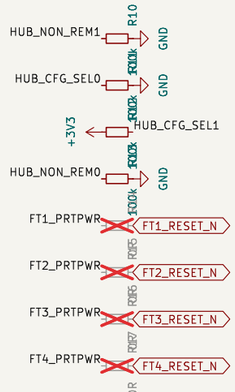
Hub port 1..4 -> FT3, FT4, FT2, FT1 (uhubctl -p N).

PRTPWR -> FT RESET# OR links are DNP: fit them and remove the FT RESET# pull-up for hard reset.









SBC farm

Sheet: /Hub/

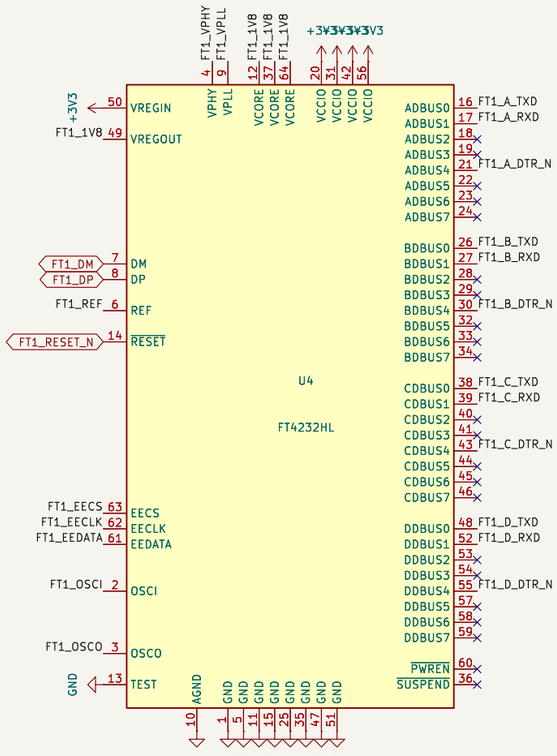
File: hub.kicad_sch

Title: USB2514B 4-port hub

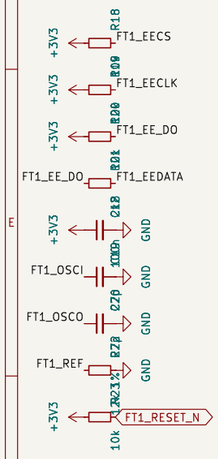
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 3/7

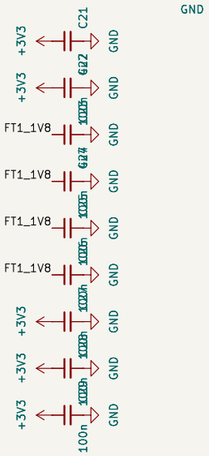
FT4232H #1 –> CH01..CH04. EEPROM serial: SBCU-1 (program with ftdi_eeprom).

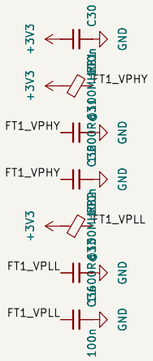
TX is driven only while DTR# is asserted (port open). LVC Ioff => Hi-Z when unpowered.

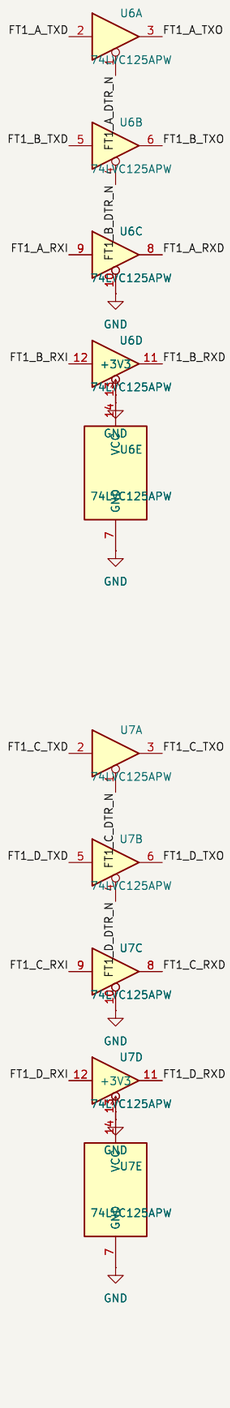


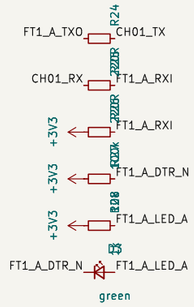


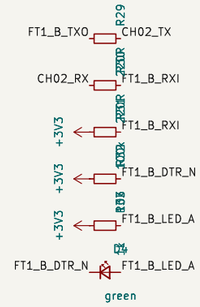


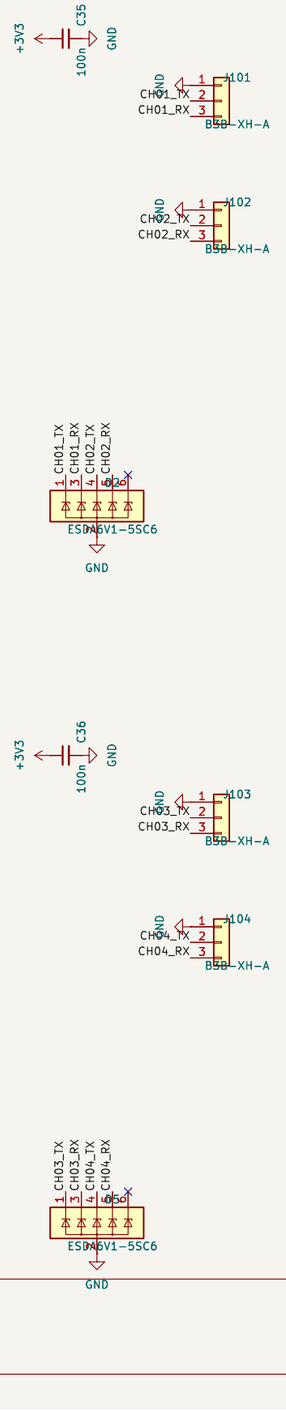


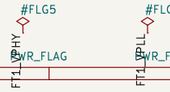












SBC farm

Sheet: /FT1/

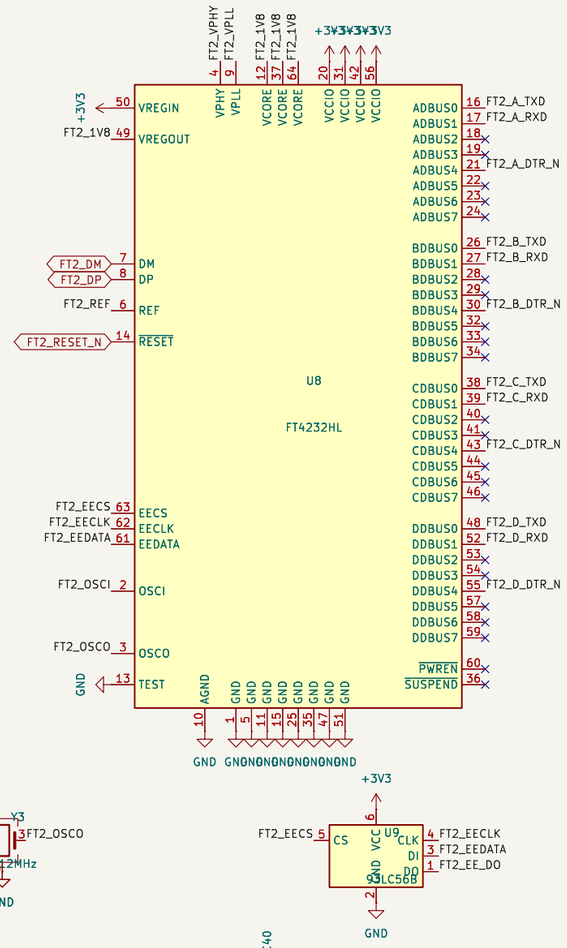
File: ft1.kicad_sch

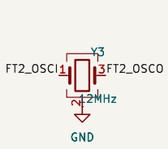
Title: FT4232H #1: channels 01–04

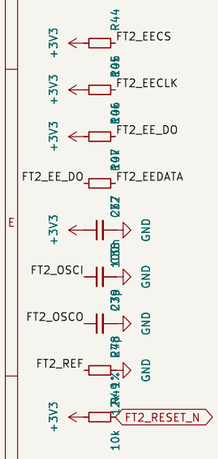
Size: A3	Date:	Rev: A
KiCad E.D.A. 9.0.9		Id: 4/7

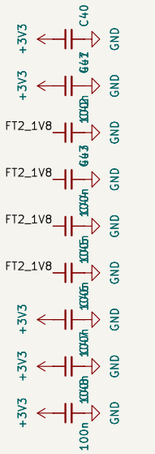
FT4232H #2 -> CH05..CH08. EEPROM serial: SBCU-2 (program with ftdi_eeprom).

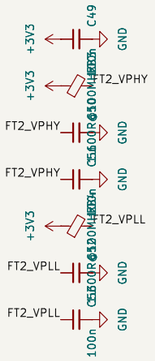
TX is driven only while DTR# is asserted (port open). LVC Ioff => Hi-Z when unpowered.

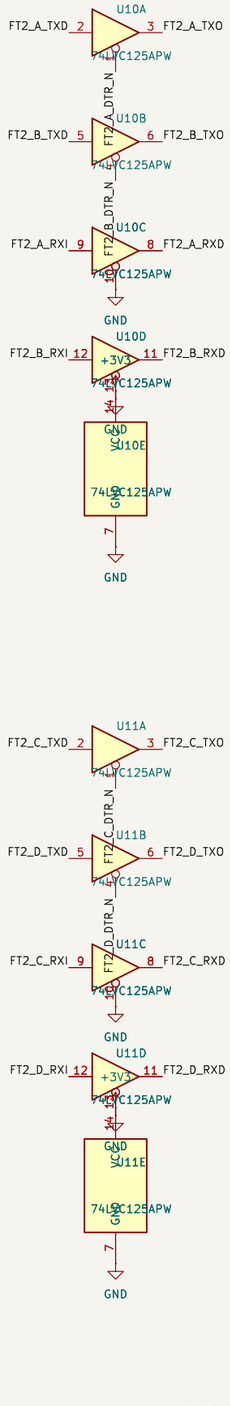


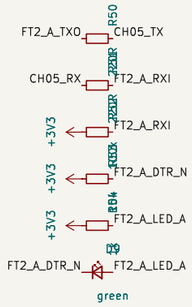


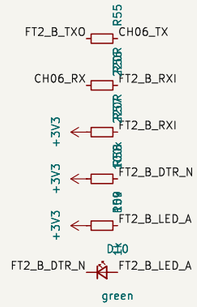


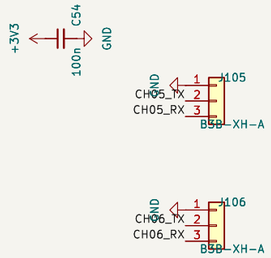


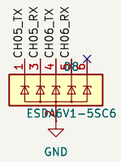


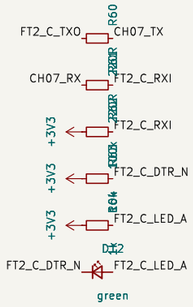


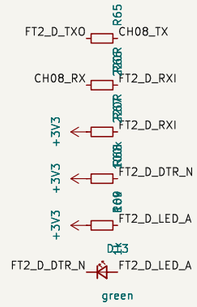


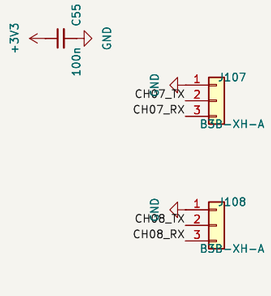


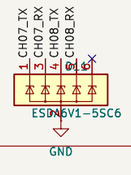


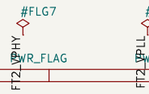












SBC farm

Sheet: /FT2/

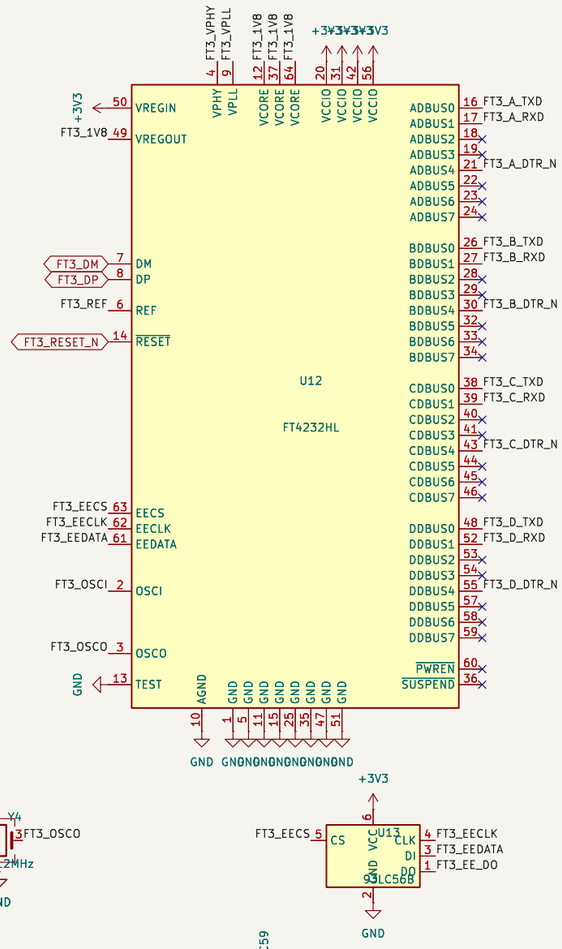
File: ft2.kicad_sch

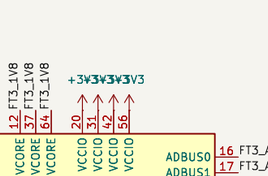
Title: FT4232H #2: channels 05–08

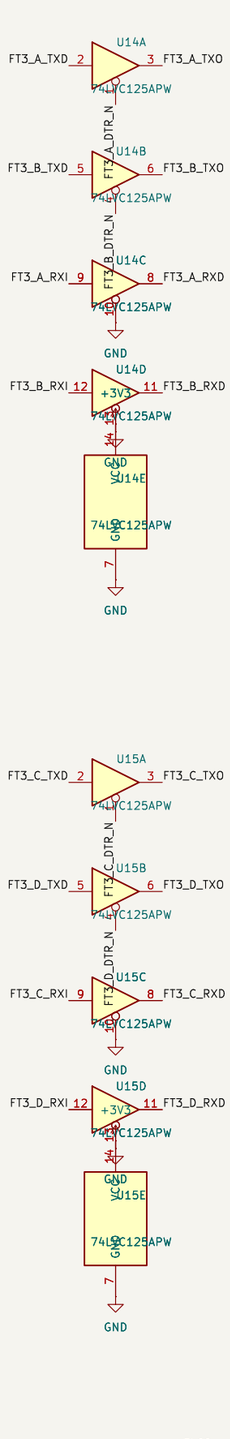
Size: A3	Date:	Rev: A
KiCad E.D.A. 9.0.9		Id: 5/7

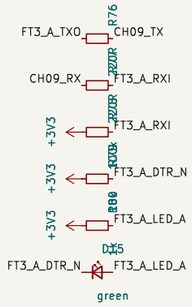
FT4232H #3 -> CH09..CH12. EEPROM serial: SBCU-3 (program with ftdi_eeprom).

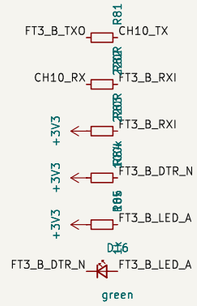
TX is driven only while DTR# is asserted (port open). LVC Ioff => Hi-Z when unpowered.

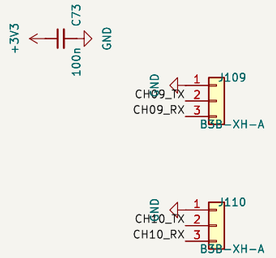


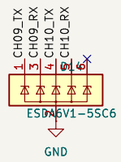


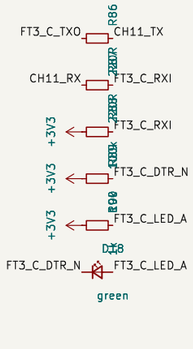


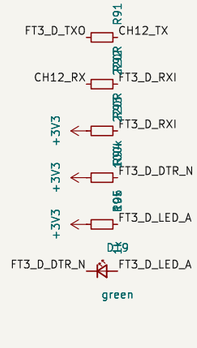


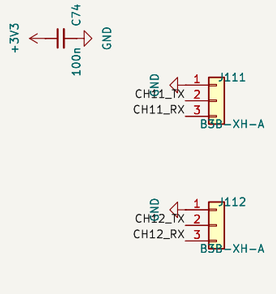


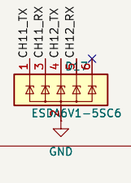


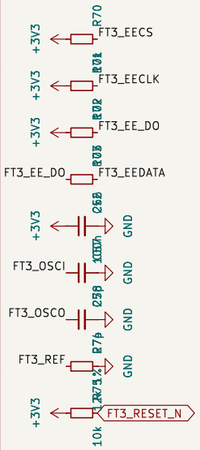


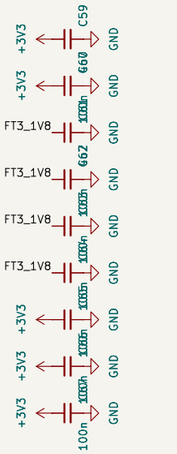


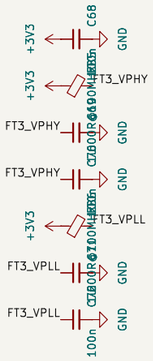
















SBC farm

Sheet: /FT3/

File: ft3.kicad_sch

Title: FT4232H #3: channels 09-12

Size: A3 | Date:

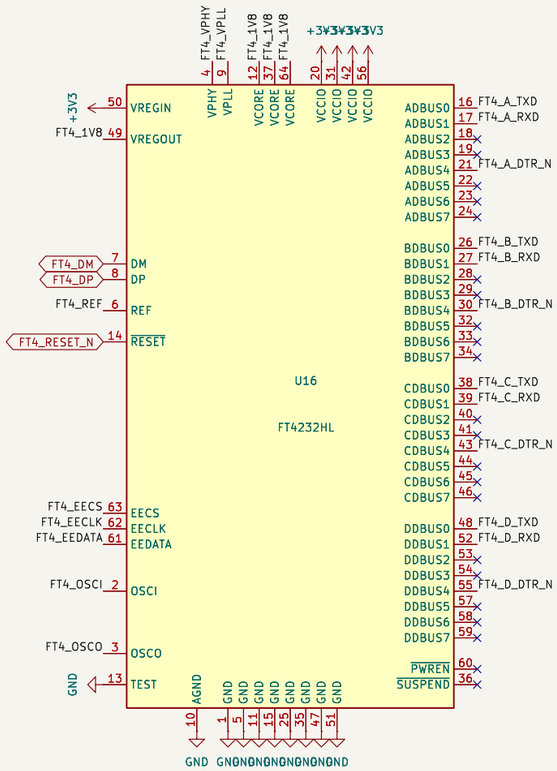
KiCad E.D.A. 9.0.9

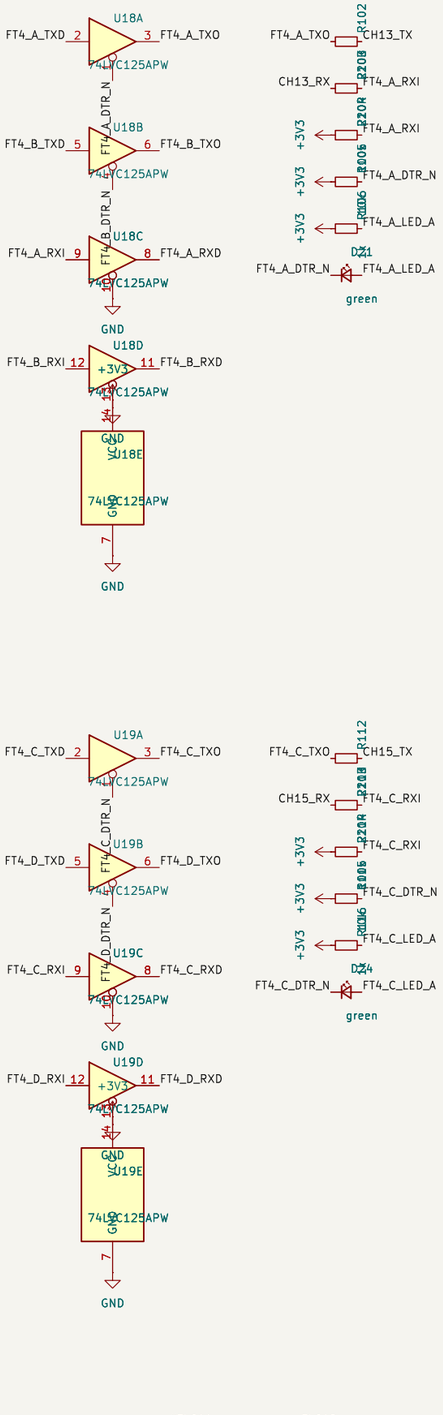
Rev: A

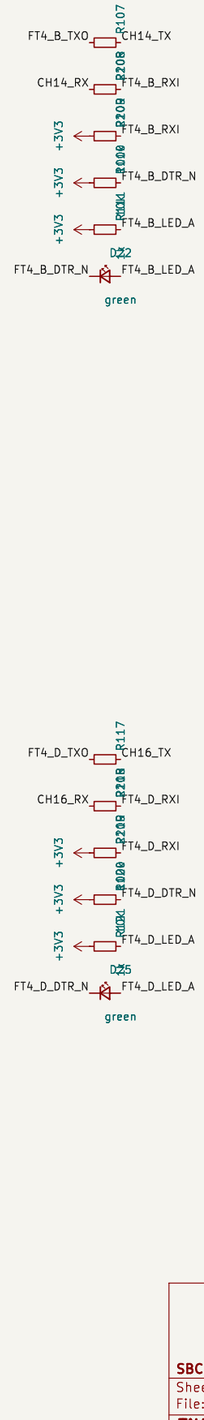
Id: 6/7

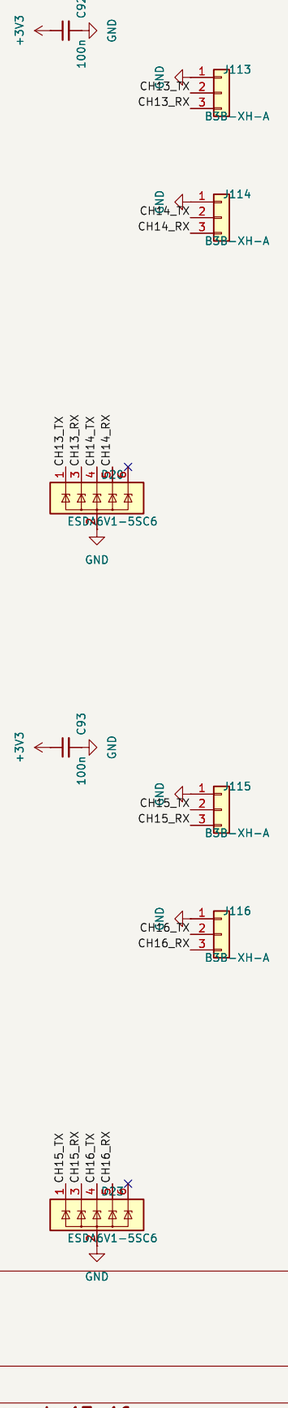
FT4232H #4 –> CH13..CH16. EEPROM serial: SBCU-4 (program with ftdi_eeprom).

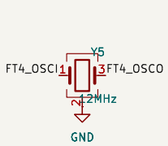
TX is driven only while DTR# is asserted (port open). LVC Ioff => Hi-Z when unpowered.

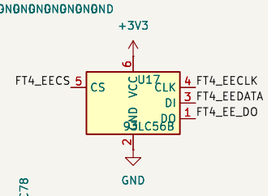


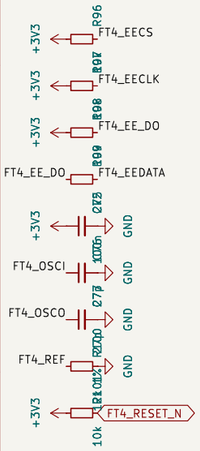












#FLG11 #FLG12

FT4_VPHY FT4_VPLL

PWR_FLAG PWR_FLAG

SBC farm

Sheet: /FT4/

File: ft4.kicad_sch

Title: FT4232H #4: channels 13–16

Size: A3	Date:	Rev: A
KiCad E.D.A. 9.0.9		Id: 7/7